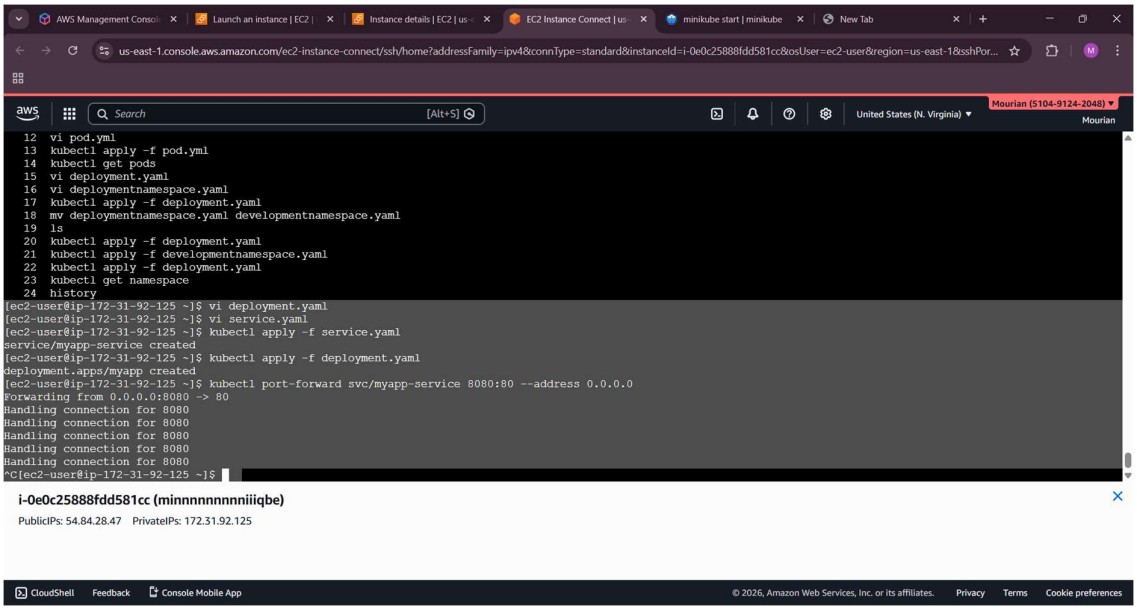
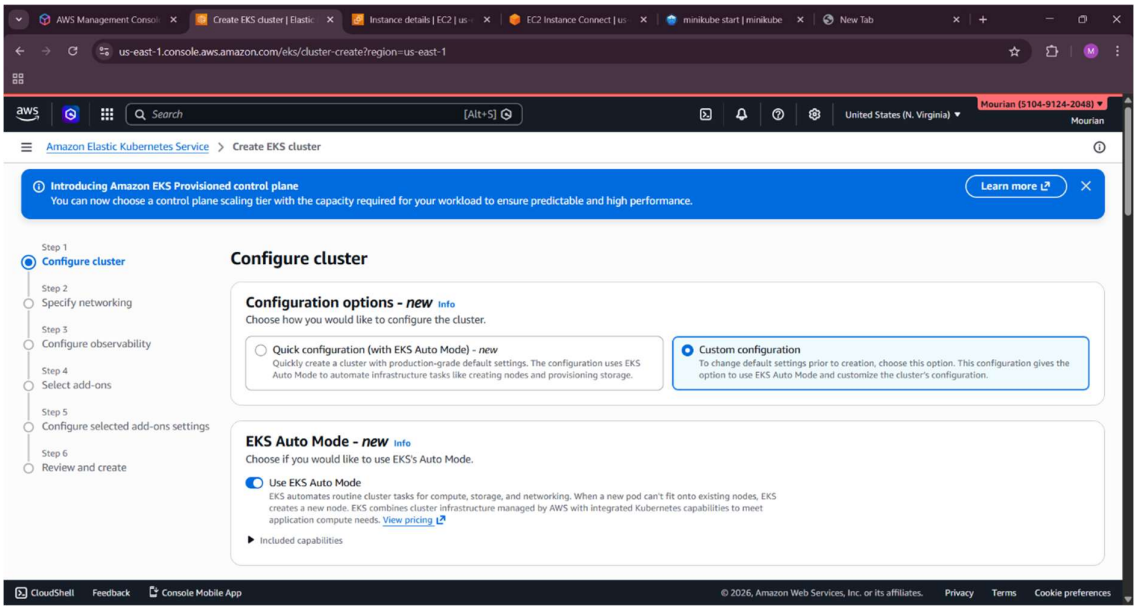
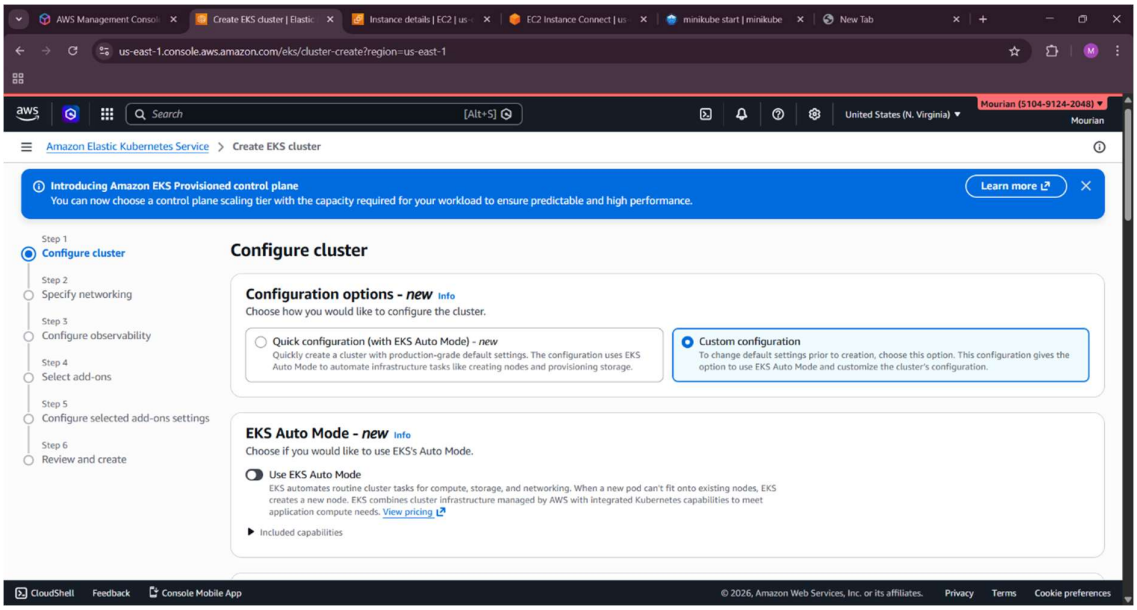
ASSIGNMENT 40

kubernetes- node group and nodeport



EKS Elastic Kubernetes:




us-east-1.console.aws.amazon.com/eks/cluster-create?region=us-east-1
Amazon Elastic Kubernetes Service > Create EKS cluster
Introducing Amazon EKS Provisioned control plane
You can now choose a control plane scaling tier with the capacity required for your workload to ensure predictable and high performance.
Learn more
Step 1
Configure cluster
Step 2
Specify networking
Step 3
Configure observability
Step 4
Select add-ons
Step 5
Configure selected add-ons settings
Step 6
Review and create
Configure cluster
Configuration options - new Info
Choose how you would like to configure the cluster.
Quick configuration (with EKS Auto Mode) - new
Quickly create a cluster with production-grade default settings. The configuration uses EKS Auto Mode to automate infrastructure tasks like creating nodes and provisioning storage.
Custom configuration
To change default settings prior to creation, choose this option. This configuration gives the option to use EKS Auto Mode and customize the cluster's configuration.
EKS Auto Mode - new Info
Choose if you would like to use EKS's Auto Mode.
Use EKS Auto Mode
EKS automates routine cluster tasks for compute, storage, and networking. When a new pod can't fit onto existing nodes, EKS creates a new node. EKS combines cluster infrastructure managed by AWS with integrated Kubernetes capabilities to meet application compute needs. View pricing
Included capabilities
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Console Mobile App
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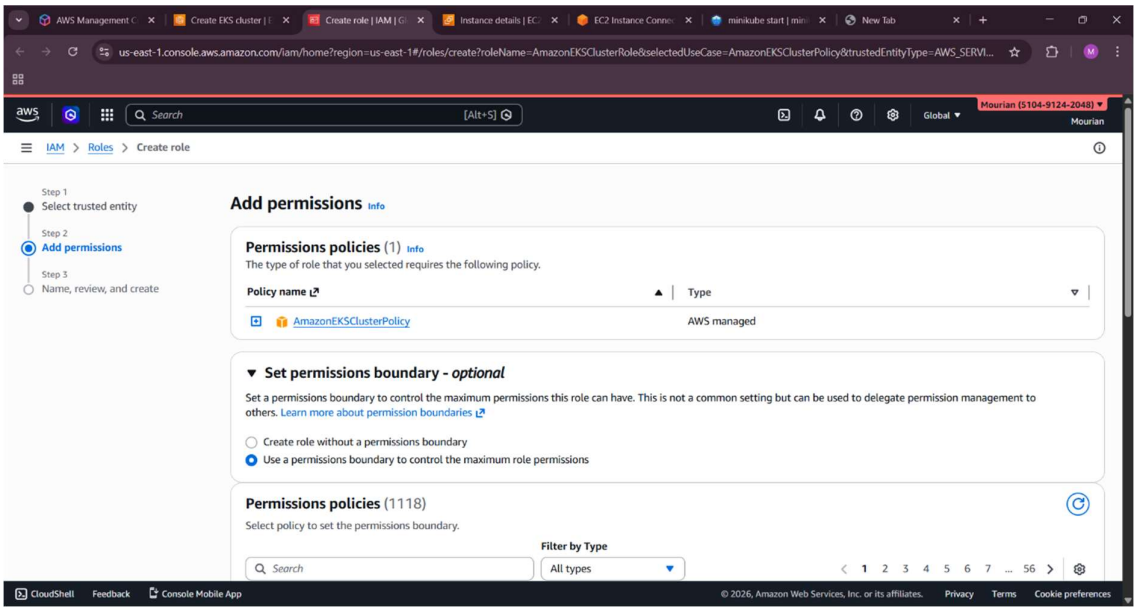
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create?roleName=AmazonEKSClusterRole&selectedUseCase=AmazonEKSClusterPolicy&trustedEntityType=AWS_SERVI...
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (1) Info
The type of role that you selected requires the following policy.
Policy name
Type
AmazonEKSClusterPolicy
AWS managed
Set permissions boundary - optional
Set a permissions boundary to control the maximum permissions this role can have. This is not a common setting but can be used to delegate permission management to others. Learn more about permission boundaries
Create role without a permissions boundary
Use a permissions boundary to control the maximum role permissions
Permissions policies (1118)
Select policy to set the permissions boundary.
Filter by Type
All types
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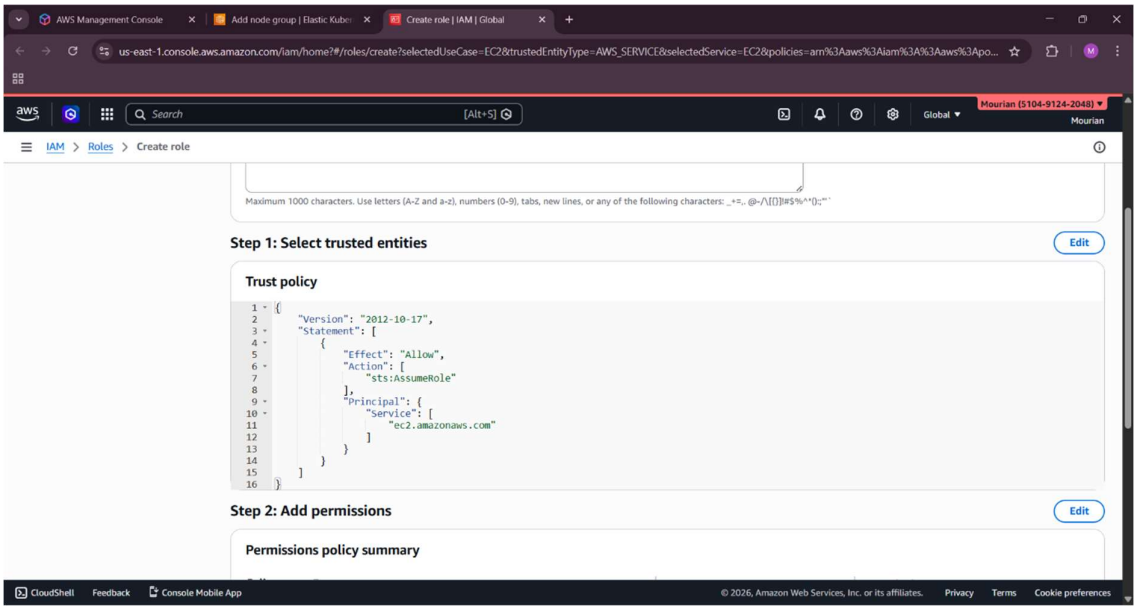
AWS Management Console
Add node group | Elastic Kube
Create role | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?#/roles/create?selectedUseCase=EC2&trustedEntityType=AWS_SERVICE&selectedService=EC2&policies=arm%3Aaws%3Aiam%3A%3Aaws%3Apo...
Search
[Alt+S]
Global
Mourian (5104-9124-2048)
Mourian
IAM > Roles > Create role
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,.@-/\[{}]!#$%^*();:"'
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
"sts:AssumeRole"
],
"Principal": {
"Service": [
"ec2.amazonaws.com"
]
}
}
]
}
Step 2: Add permissions
Edit
Permissions policy summary
CloudShell
Feedback
Console Mobile App
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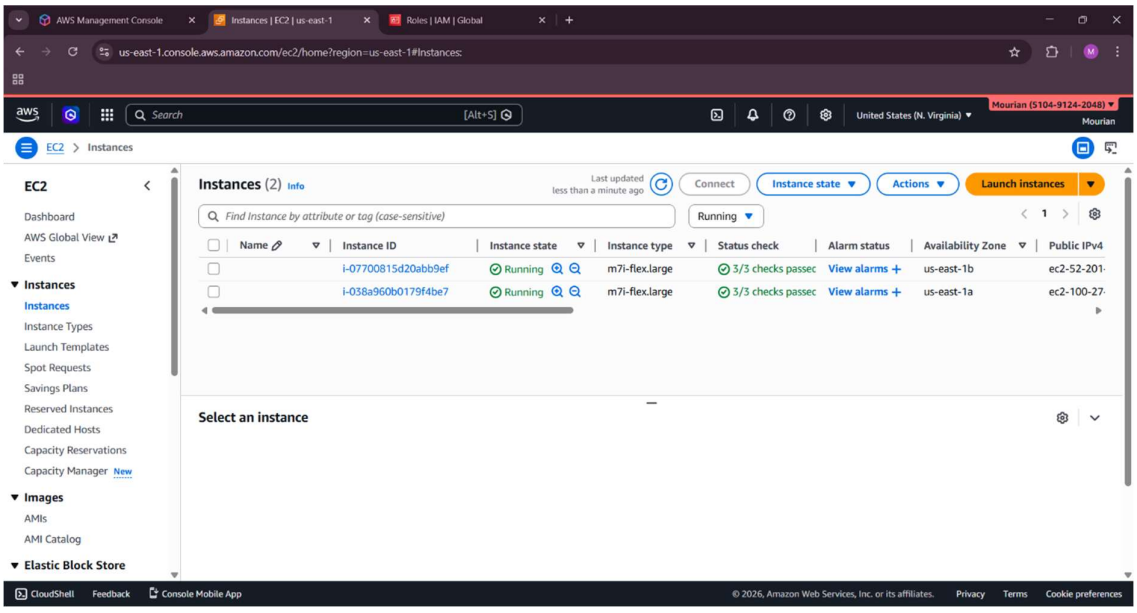
AWS Management Console
Instances | EC2 | us-east-1
Roles | IAM | Global
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search
[Alt+S]
United States (N. Virginia)
Mourian (5104-9124-2048)
Mourian
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instances (2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
Running
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
i-07700815d20abb9ef
Running
m7i-flex.large
3/3 checks passed
View alarms +
us-east-1b
ec2-52-201-
i-038a960b0179f4be7
Running
m7i-flex.large
3/3 checks passed
View alarms +
us-east-1a
ec2-100-27-
Select an instance
CloudShell
Feedback
Console Mobile App
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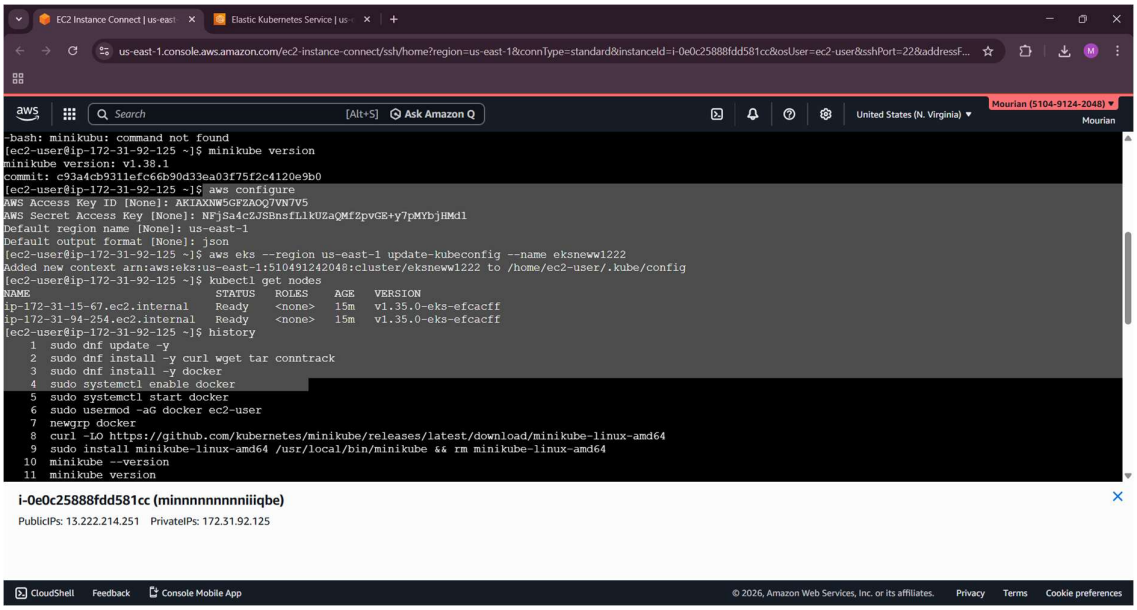
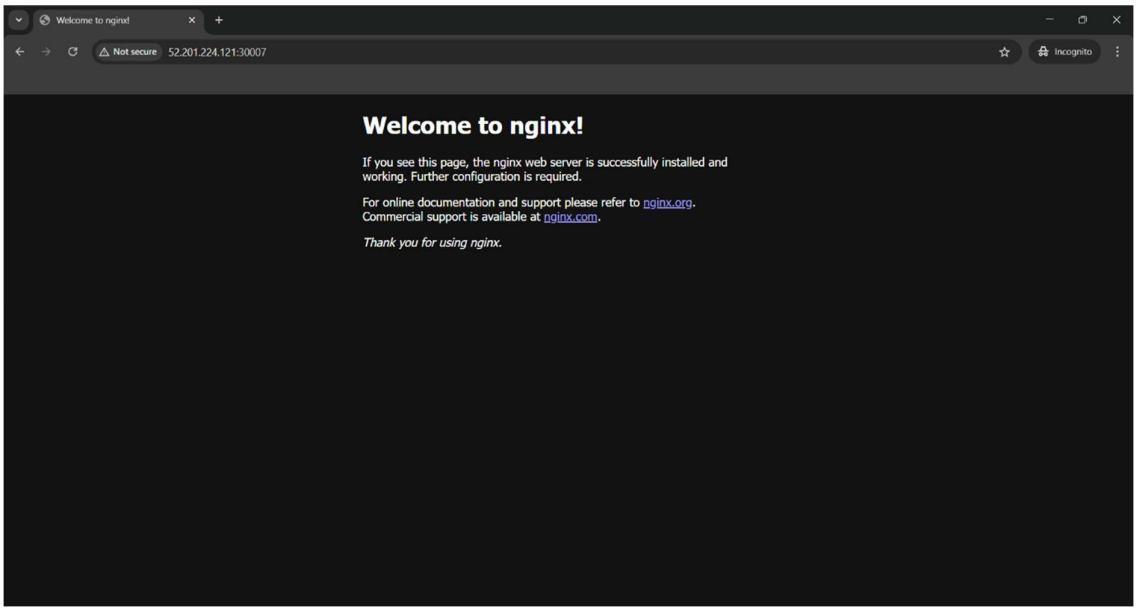


This output from EKS two instance nginx web hosting by using node port :



Deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.25-alpine
          ports:
            - containerPort: 80
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "200m"
              memory: "256Mi"
```

Service.yaml:

```yaml
apiVersion: v1
```

```yaml
kind: Service
metadata:
  name: nginx-nodeport-service
  labels:
    app: nginx
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007   # Must be between 30000-32767
```